

Empirical Evidence on Quantitative Easing: Effects Across Major Economies

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Abstract

This document summarizes the empirical literature on the macroeconomic effects of quantitative easing (QE) programs implemented by major central banks since the 2008 Global Financial Crisis. We review evidence on effects on bond yields, GDP, and inflation across the United States, United Kingdom, Euro Area, Japan, and Sweden. A dedicated section examines Sweden's experience, drawing on the comprehensive 2026 Riksbank evaluation. The evidence suggests QE reduced long-term yields by 40–150 basis points, raised GDP by 1–3%, and increased inflation by 0.3–2 percentage points, with substantial variation across countries and methodologies.

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1 Introduction

Quantitative easing (QE) refers to large-scale asset purchases by central banks, typically of government bonds, financed by the creation of central bank reserves. QE became a primary monetary policy tool when conventional interest rate policy reached the zero lower bound following the 2008 Global Financial Crisis.

This summary draws on approximately 50 source documents including Federal Reserve speeches, central bank working papers, and academic research to synthesize the empirical evidence on QE effectiveness.

2 Transmission Channels

QE affects the economy through several channels:

1. **Portfolio Balance:** Central bank purchases remove duration risk from private portfolios, lowering term premia and pushing investors toward riskier assets.
2. **Signaling:** Asset purchases signal commitment to maintaining low policy rates, reducing expected future short rates.
3. **Liquidity:** During market stress, central bank purchases improve market functioning and reduce liquidity premia.
4. **Exchange Rate:** Lower domestic yields reduce currency attractiveness, potentially causing depreciation and boosting net exports.
5. **Bank Capital:** Rising bond prices improve bank capital positions, enabling increased lending.

The relative importance of each channel varies by country and economic context.

3 Effects on Bond Yields

The most directly measurable effect of QE is on government bond yields. Table 1 summarizes event study estimates across major programs.

Table 1: Estimated Effects of QE on 10-Year Government Bond Yields

Country	Program	Yield Impact	Source
4*United States	QE1 (\$1.7T)	−40 to −110 bps	Bernanke (2012)
	QE1	−107 bps	Krishnamurthy & V-J (2011)
	QE2 (\$600B)	−15 to −45 bps	Bernanke (2012)
3*United Kingdom	QE1 (£200B)	−100 bps	Joyce et al. (2011)
	QE1/QE2	−47 bps	Christensen & Rudebusch (2012)
	QE1	−150 bps	Bridges & Thomas (2012)
3*Euro Area	APP	−45 bps (avg)	Andrade et al. (2016)
	OMT	−199 bps (Italy 2y)	Altavilla et al. (2014)
	SMP	−32 to −40 bps	De Pooter et al. (2015)
2*Japan	QQE1	−10 to −14 bps	Various

3.1 Key Findings

- Effects were larger during periods of financial stress (QE1 vs. later rounds)
- Portfolio balance effects dominated over signaling effects in most studies
- Japanese effects were smaller due to already-low initial yields
- Euro area peripheral countries experienced much larger yield reductions

4 Effects on GDP

Table 2 summarizes estimates of QE effects on real GDP.

Table 2: Estimated Effects of QE on GDP (Peak Effect)

Country	Study	Method	GDP Impact
2*United States	Bernanke (2012)	Fed models	+3%
	Baumeister & Benati (2013)	TVP-VAR	Prevented –10% collapse
4*United Kingdom	Bridges & Thomas (2012)	Monetarist	+2% peak
	Kapetanios et al. (2012)	VAR	+1.5%
	Baumeister & Benati (2013)	TVP-VAR	+3%
	Weale & Wieladek (2016)	VAR	+0.25% per 1% GDP
3*Euro Area	Andrade et al. (2016)	APP	+1.1%
	Altavilla et al. (2014)	OMT	+1.5% (Italy)
	Hutchinson & Smets (2017)	Combined	+1.7% (2016–19)
2*Japan	Hausman & Wieland (2014)	QQE1	up to +1%
	Kan et al. (2016)	QQE1	+0.6% to +4.2%

4.1 Counterfactual Analysis

Baumeister and Benati (2013) provide particularly striking counterfactual estimates using time-varying parameter VARs:

- **United States:** Without QE, GDP growth would have contracted by 10% (annualized) in 2009Q1, with unemployment exceeding 10.6%
- **United Kingdom:** Output growth would have dropped by 12% at annual rates
- Both economies faced “output collapses comparable to those that took place during the Great Depression”

5 Effects on Inflation

QE effects on inflation were generally more modest than effects on output. Table 3 summarizes the evidence.

Table 3: Estimated Effects of QE on Inflation

Country	Study	Inflation Impact
2*United States	Baumeister & Benati (2013)	Prevented -1% deflation
3*United Kingdom	Bridges & Thomas (2012)	+1.0 pp
	Kapetanios et al. (2012)	+1.25 pp
	Baumeister & Benati (2013)	+2.0 pp
3*Euro Area	Andrade et al. (2016)	+0.4 pp (actual), +0.45 pp (expectations)
	Altavilla et al. (2014)	+1.2 pp (Italy)
	Hutchinson & Smets (2017)	+0.5 pp (cumulative 2016–19)
2*Japan	Kan et al. (2016)	+0.3 to +1.5 pp
	De Michelis & Iacoviello (2016)	+0.8 pp

5.1 Why No Hyperinflation?

Despite massive balance sheet expansion, inflation remained subdued because:

- Large output gaps meant little resource pressure
- Banks held excess reserves rather than expanding lending proportionally
- Long-term inflation expectations remained anchored near targets
- Global disinflationary forces (technology, globalization) persisted

6 Cross-Country Comparison

Table 4 compares QE programs across major economies.

Table 4: Cross-Country QE Comparison

	US	UK	Euro Area	Japan
Peak Balance Sheet (% GDP)	~35%	~45%	~60%	>100%
Negative Rates	No	No	Yes (-0.50%)	Yes (-0.10%)
Yield Curve Control	No	No	No	Yes
GDP Effect (typical)	2–3%	1.5–2%	1–1.5%	0.5–1%
Inflation Effect	Modest	+1–2 pp	+0.5 pp	Limited

6.1 Factors Explaining Variation

1. **Initial Conditions:** Higher starting yields allow larger compression
2. **Financial Stress:** Effects larger during market dysfunction
3. **Central Bank Credibility:** Japan’s deflationary history limited effectiveness
4. **Economic Structure:** Bank-dependent economies respond differently
5. **Exchange Rate:** Small open economies see larger exchange rate effects

7 Sweden: A Comprehensive Case Study

Sweden provides a distinctive case study due to its small, open economy and the availability of a comprehensive independent evaluation (Ravn & Wilkins, 2026) commissioned by the Swedish Parliament.

7.1 Timeline of Unconventional Policies

Table 5: Sweden’s Unconventional Monetary Policy Timeline

Date	Action	Details
February 2015	First negative rate	Policy rate to -0.10%
February 2015	QE begins	SEK 10 billion government bonds
February 2016	Lowest rate	Policy rate reaches -0.50%
2015–2017	QE expansion	Holdings reach SEK 330 billion
December 2019	Exit negative rates	Policy rate returns to 0%
March 2020	Covid response	SEK 300 billion program announced
November 2020	Expansion	Envelope increased to SEK 700 billion
Early 2022	Peak holdings	$\sim$ SEK 1 trillion ($\sim 25\%$ of GDP)
April 2022	Rate hikes begin	First increase to 0.25%
September 2024	Capital injection	SEK 25 billion from government

7.2 Macroeconomic Effects in Sweden

Table 6 presents estimates of QE effects in Sweden.

Table 6: Estimated QE Effects in Sweden

Period	Source	GDP Effect	Inflation Effect
2015–2019	Kolasa, Laséen & Lindé (2025)	$\sim 1\%$	+20–50 bps
2020–2023	Akkaya et al. (2023)	$\sim 0.2\%$	+0.25 pp

Key finding: Pre-Covid QE was substantially more effective than Covid-era QE. The weaker effects during 2020–2023 reflect:

- Asset composition weighted toward covered bonds (weaker transmission)
- Market stress had already subsided when some programs were implemented
- Smaller purchases relative to the economy

7.3 Transmission Channels in Sweden

The **exchange rate channel** was particularly important for Sweden as a small, open economy:

“Kolasa, Laséen & Lindé (2025) find QE was effective largely because it led to real depreciation of the krona, stimulating exports and imported inflation.” (Ravn & Wilkins, 2026)

Additional Sweden-specific channels:

- **Cash flow channel:** Strong due to predominance of variable-rate mortgages
- **Signaling:** Forward guidance effective; Riksbank published policy rate paths since 2007

7.4 Negative Interest Rates in Sweden

Sweden maintained negative rates from February 2015 to December 2019, reaching -0.50% .

Table 7: Pass-Through of Negative Rates in Sweden

Rate	Pass-Through Coefficient
Interbank rate (STIBOR)	~ 1.0
3-month mortgage rate	~ 1.0
Corporate loan rate	~ 1.0
Deposit rate	Substantial
10-year government bond	~ 0.1

Diminishing returns: Eggertsson et al. (2024) find that pass-through to lending rates declined when the policy rate fell below -0.25% , suggesting Sweden may have approached the “reversal rate” where further cuts become counterproductive.

7.5 Balance Sheet Consequences

Sweden’s experience highlights the fiscal costs of QE:

- **Peak balance sheet:** 27.5% of GDP (early 2022)
- **Maturity mismatch:** Long-term bond assets vs. short-term liabilities
- **Mark-to-market losses:** When rates rose in 2022
- **Capital injection:** SEK 25 billion (0.39% of GDP) in September 2024

The Riksbank had requested SEK 43.7 billion but received a smaller amount.

7.6 Monetary-Fiscal Coordination

Sweden operated under strict monetary dominance with:

- Independent Riksbank pursuing 2% inflation target
- Fiscal rules: surplus target, expenditure ceiling, 35% debt anchor

This meant the burden of stabilization at the zero lower bound fell entirely on the Riksbank. The National Debt Office continued issuing longer-term government bonds throughout the QE period, which the Riksbank then purchased—creating interest rate risk concentration.

7.7 The Role of Wage-Setting

A unique Swedish factor: the **Industrial Agreement** (Industriavtalet) from 1997 establishes coordinated wage bargaining with $\sim 90\%$ of employees covered by collective agreements.

During 2022–2023, wage increases were moderate ($\sim 3.7\%$ annually) despite high inflation, contributing to real wage declines but helping anchor inflation expectations. This dampened the risk of wage-price spirals.

7.8 Lessons from Sweden

The 2026 Ravn-Wilkins evaluation identifies key lessons:

1. **Distinguish market-functioning from stimulus QE:** The former was highly effective during Covid; the latter had modest effects
2. **Exchange rate channel dominates** for small, open economies
3. **Balance sheet risks** should be assessed more systematically ex-ante
4. **Exit strategies** should be predefined
5. **Coordination** between monetary and fiscal authorities matters, even within a monetary dominance framework

8 Methodological Considerations

8.1 Why Estimates Vary

Fabo et al. (2021) document that central bank researchers systematically report larger QE effects than academic researchers:

- Gap persists after controlling for methodology
- May reflect institutional incentives, selection effects, or data access
- Raises concerns about research credibility

8.2 Comparison of Methods

Table 8: Methodological Approaches to Measuring QE Effects

Method	Strengths	Limitations
Event Studies	Clean identification	Captures only surprises
VAR	Dynamic effects, GDP/inflation	Identification challenges
TVP-VAR	Allows time variation	Complex, small samples
DSGE	Theory-consistent	Model-dependent
Narrative	Natural experiments	Rare events

8.3 Swedish Methodological Comparison

Even for the same country, different methods yield substantially different estimates:

Table 9: Swedish Monetary Policy Estimates by Method

Study	Method	GDP Effect (1 pp rate)
Berggren et al. (2024)	Bayesian VAR	−0.7% to −0.8%
Almerud et al. (2024)	High-frequency ID	−1.5%
Coglianese et al. (2025)	Narrative	−5%

9 Conclusion

The empirical evidence supports several conclusions:

1. **QE reduced long-term yields** by 40–150 basis points, with effects concentrated during financial stress
2. **QE raised GDP** by 1–3% at peak, with counterfactual analysis suggesting it prevented much worse outcomes
3. **Inflation effects were modest** (0.3–2 pp), preventing deflation rather than causing high inflation
4. **Effects varied substantially** across countries due to initial conditions, credibility, and economic structure
5. **Sweden’s experience** demonstrates the importance of the exchange rate channel for small open economies and highlights balance sheet risks that materialized when rates rose
6. **Methodological uncertainty** remains large, with estimates varying by factors of 3–10x depending on approach

The Swedish case, with its comprehensive 2026 evaluation, provides particularly valuable lessons for future unconventional monetary policy design.

Key Sources

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